

EECS 498/CSE 598: Zero-Knowledge Proofs

Winter 2026

Lecture 23

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Agenda for this lecture

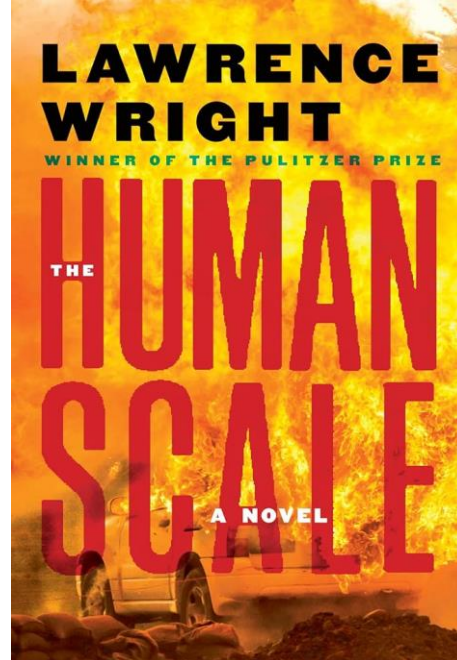
- Announcements
- (Finish Zerocash, then..)Background on anonymous credentials
- Overview of different anonymous credentials constructions
- Deep dive: keyed-verification anonymous credentials
- Legacy-compatible anonymous credentials via zkSNARKs
- Use cases, deployments, controversies

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Announcements

- P4 and theory pset 2 are out!.
 - Note there will be no P5 nor a final exam. These will be the last assignments of the semester.
- All work for this course must be completed by our final exam date, April 27th
- Reminder: you can get participation credit for going to discussion.
- If something is stated clearly in the syllabus, and you ask me about it on Piazza instead of just looking at the syllabus to answer your question first, it makes me sad.
- I read a good book recently



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The NP Statement POUR

Instances have the form

$$x = (\text{rt}, \text{sn}^{\text{old}}, \text{cm}_1^{\text{new}}, \text{cm}_2^{\text{new}}, v_{\text{pub}}).$$

Witnesses have the form

$$w = (\text{path}, c^{\text{old}}, a_{\text{sk}}^{\text{old}}, c_1^{\text{new}}, c_2^{\text{new}}),$$

where $c^{\text{old}} = (a_{\text{pk}}^{\text{old}}, v^{\text{old}}, \rho^{\text{old}}, r^{\text{old}}, s^{\text{old}}, \text{cm}^{\text{old}})$ and each $c_i^{\text{new}} = (a_{\text{pk},i}^{\text{new}}, v_i^{\text{new}}, \rho_i^{\text{new}}, r_i^{\text{new}}, s_i^{\text{new}}, \text{cm}_i^{\text{new}})$.

A witness w is *valid* for instance x if the following hold:

1. **Membership.** path is a valid authentication path for leaf cm^{old} with respect to root rt in a CRH-based Merkle tree.
2. **Ownership.** $a_{\text{pk}}^{\text{old}} = \text{PRF}_{a_{\text{sk}}^{\text{old}}}^{\text{addr}}(0)$.
3. **Serial number correctness.** $\text{sn}^{\text{old}} = \text{PRF}_{a_{\text{sk}}^{\text{old}}}^{\text{sn}}(\rho^{\text{old}})$.
4. **Old coin well-formedness.** $\text{cm}^{\text{old}} = \text{COMM}_{s^{\text{old}}}(v^{\text{old}} \parallel \text{COMM}_{r^{\text{old}}}(a_{\text{pk}}^{\text{old}} \parallel \rho^{\text{old}}))$.
5. **New coin well-formedness.** For each $i \in \{1, 2\}$, $\text{cm}_i^{\text{new}} = \text{COMM}_{s_i^{\text{new}}}(v_i^{\text{new}} \parallel \text{COMM}_{r_i^{\text{new}}}(a_{\text{pk},i}^{\text{new}} \parallel \rho_i^{\text{new}}))$.
6. **Balance.** $v_1^{\text{new}} + v_2^{\text{new}} + v_{\text{pub}} = v^{\text{old}}$, with $v^{\text{old}} \leq v_{\text{max}}$.

The Pour Algorithm

Inputs:

- Public parameters $\text{pp} = (\text{pk}_{\text{POUR}}, \text{vk}_{\text{POUR}})$
- Merkle root rt
- Old coin c^{old} with address secret key $a_{\text{sk}}^{\text{old}}$
- Authentication path path
- New values $v_1^{\text{new}}, v_2^{\text{new}}$ and destination public keys $a_{\text{pk},1}^{\text{new}}, a_{\text{pk},2}^{\text{new}}$
- Public output v_{pub} and transaction string info

Procedure:

1. Parse $c^{\text{old}} = (a_{\text{pk}}^{\text{old}}, v^{\text{old}}, \rho^{\text{old}}, r^{\text{old}}, s^{\text{old}}, \text{cm}^{\text{old}})$ and compute

$$\text{sn}^{\text{old}} := \text{PRF}_{a_{\text{sk}}^{\text{old}}}^{\text{sn}}(\rho^{\text{old}}).$$

2. For each $i \in \{1, 2\}$, sample fresh $\rho_i^{\text{new}} \leftarrow \{0, 1\}^\lambda$ and commitment randomness $r_i^{\text{new}}, s_i^{\text{new}}$, then compute

$$\begin{aligned} k_i^{\text{new}} &:= \text{COMM}_{r_i^{\text{new}}}(a_{\text{pk},i}^{\text{new}} \parallel \rho_i^{\text{new}}), \\ \text{cm}_i^{\text{new}} &:= \text{COMM}_{s_i^{\text{new}}}(v_i^{\text{new}} \parallel k_i^{\text{new}}). \end{aligned}$$

$$\text{Set } c_i^{\text{new}} := (a_{\text{pk},i}^{\text{new}}, v_i^{\text{new}}, \rho_i^{\text{new}}, r_i^{\text{new}}, s_i^{\text{new}}, \text{cm}_i^{\text{new}}).$$

3. Form the SNARK instance and witness:

$$\begin{aligned} x &:= (\text{rt}, \text{sn}^{\text{old}}, \text{cm}_1^{\text{new}}, \text{cm}_2^{\text{new}}, v_{\text{pub}}), \\ w &:= (\text{path}, c^{\text{old}}, a_{\text{sk}}^{\text{old}}, c_1^{\text{new}}, c_2^{\text{new}}). \end{aligned}$$

4. Compute $\pi \leftarrow \text{Prove}(\text{pk}_{\text{POUR}}, x, w)$.
5. Output new coins $c_1^{\text{new}}, c_2^{\text{new}}$ and the pour transaction

$$\text{tx}_{\text{pour}} := (\text{rt}, \text{sn}^{\text{old}}, \text{cm}_1^{\text{new}}, \text{cm}_2^{\text{new}}, v_{\text{pub}}, \text{info}, \pi).$$

v_{pub} and *info* never bound to π
can just be stripped off and replaced
with another address!

Even if bound, need stronger non-malleability property of NIZK to prevent tampering provably.

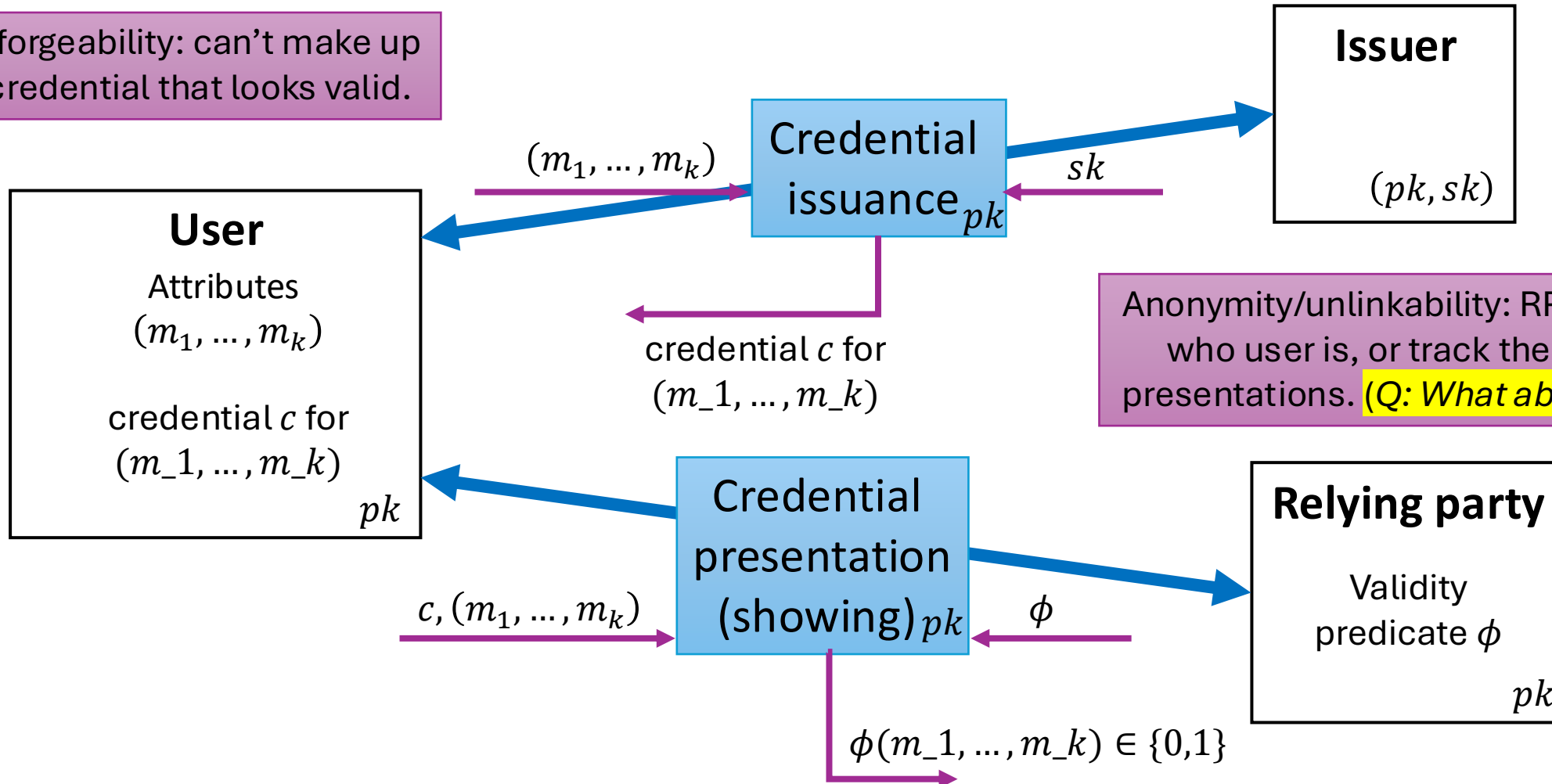
Simulation-soundness and simulation-extractibility

What are anonymous credentials?

Recall the “show your ID to get into a bar” scenario...

Three entities are involved: the government that issued you the ID, you, and the bouncer who wants to check your age

Unforgeability: can't make up a credential that looks valid.



Anonymity/unlinkability: RP can't learn who user is, or track them across presentations. (Q: What about issuer?)

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AC constructions

Whirlwind tour of AC work in cryptography:

- Chaum: original proposal, lacking many features
- Camenisch-Lysyanskaya – RSA-based
- U-Prove, Idemix – early AC products (Credentica/Microsoft and IBM)
- CL “template” (CL04): signatures with efficient protocols.
- Boneh-Boyen-Shacham: pairing-based signatures with efficient protocols

The “signatures with efficient protocols” recipe: Given a signature scheme where you can efficiently

1. obtain a signature on a message that’s hidden from the signer
2. prove knowledge of a signature σ on message m , without revealing σ

you can build anonymous credentials. (How?)

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Keyed-verification ACs (CMZ '14)

$$H_{\mathbf{x}}(\mathbf{m}) := x_0 + \sum_{i=1}^n x_i m_i.$$

Based on the algebraic MAC construction:

KeyGen(params): Choose and output $\mathbf{sk} := \mathbf{x} = (x_0, x_1, \dots, x_n) \leftarrow_{\$} \mathbb{F}_p^{n+1}$.
Publish issuer parameters $\mathbf{iparams} := (X_1, \dots, X_n)$ where $X_i := h^{x_i}$.

MAC(sk, m): Choose $u \leftarrow_{\$} \mathbb{G} \setminus \{1\}$. Output $\sigma = (u, u')$ where $u' := u^{H_{\mathbf{x}}(\mathbf{m})}$.

Verify(sk, m, σ): Parse $\sigma = (u, u') \in \mathbb{G}^2$. Accept iff $u \neq 1$ and $u' = u^{H_{\mathbf{x}}(\mathbf{m})}$.

Issuance (when message is known to issuer):

The issuer computes $\sigma = (u, u') \leftarrow \text{MAC}(\mathbf{sk}, \mathbf{m})$ and sends (σ, π) to the user, where π is a zero-knowledge argument of knowledge for the relation

$$\mathcal{R}_{\text{issue}} = \left\{ (u, u', \{m_i\}_{i=1}^n, C_{x_0}, \{X_i\}_{i=1}^n; x_0, \dots, x_n, \tilde{x}_0) : \begin{array}{l} u' = u^{x_0} \prod_{i=1}^n (u^{m_i})^{x_i} \\ C_{x_0} = g^{x_0} h^{\tilde{x}_0} \\ X_i = h^{x_i} \ \forall i \in \{1, \dots, n\} \end{array} \right\}.$$

The user accepts if π verifies; the credential is $\text{cred} = (u, u')$.

Keyed-verification ACs

Show $(g, h, \{X_i\}_{i=1}^n, \phi, \text{cred}, \{m_i\})$: Parse $\text{cred} = (u, u')$. Choose $r, z_1, \dots, z_n \leftarrow \mathbb{F}_p$. Compute:

$$C_{m_i} := u^{m_i} h^{z_i} \quad (i = 1, \dots, n), \quad C_{u'} := u' \cdot g^r.$$

Compute and send $\bar{\sigma} = (u, \{C_{m_i}\}, C_{u'})$ together with a zero-knowledge argument of knowledge for relation

$$\mathcal{R}_{\text{show}} = \left\{ (u, \{C_{m_i}\}, V, \{X_i\}, \phi; m_1, \dots, m_n, z_1, \dots, z_n, r) : \begin{array}{l} \phi(m_1, \dots, m_n) = 1 \\ C_{m_i} = u^{m_i} h^{z_i} \quad \forall i \\ V = g^{-r} \prod_{i=1}^n X_i^{z_i} \end{array} \right\}.$$

ShowVerify $(g, h, \text{sk}, \phi, \bar{\sigma}, \pi)$: Parse $\bar{\sigma} = (u, \{C_{m_i}\}, C_{u'})$. Compute

$$V := \frac{u^{x_0} \prod_{i=1}^n C_{m_i}^{x_i}}{C_{u'}}.$$

Verify π using V . Accept iff the proof is valid.

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zkSNARK-based ACs

Core primitive: *proof of possession* of valid (signature,message) pair for public key.

zkSNARK-based ACs

Anonymous Credentials from ECDSA

Matteo Frigo*
Google

abhi shelat†
Google

December 21, 2024

Abstract

[🏠](#) > [INNOVATION & AI](#) > [TECHNOLOGY](#) > [SAFETY & SECURITY](#)

Opening up ‘Zero-Knowledge Proof’ technology to promote privacy in age assurance

Jul 03, 2025
1 min read

Code is now available.

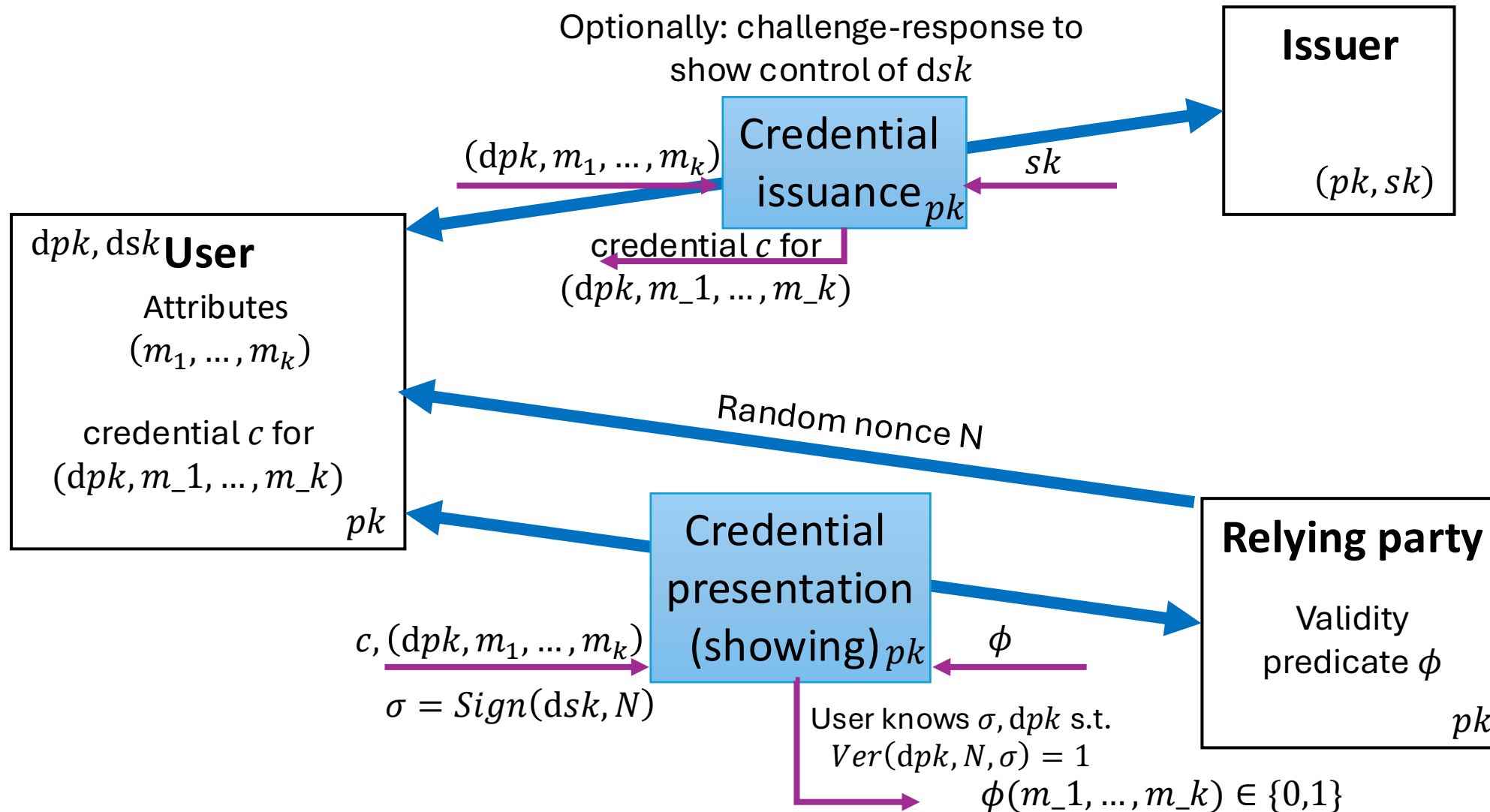


Alan Stapelberg
Group Product Manager, Google Wallet

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Device binding: augmented credential flow

Want an issued credential to be device-bound (can't be spoofed) – how to accomplish this?

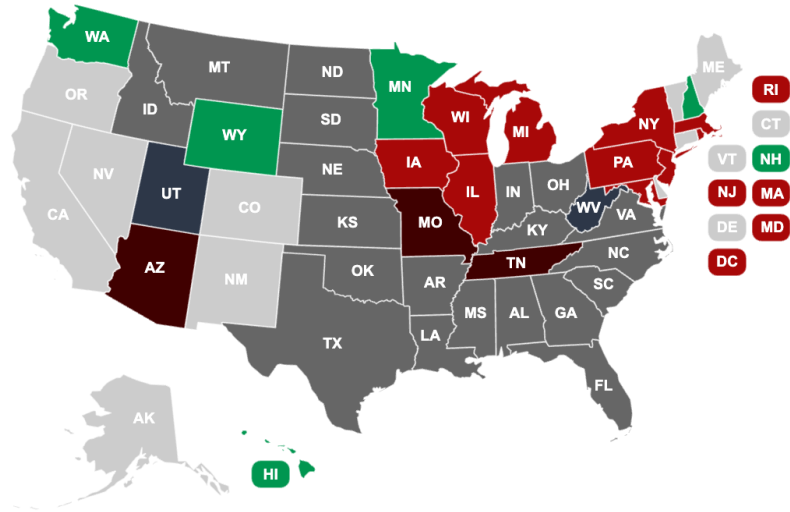


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Use cases and controversies

Age Verification Bill Tracker



KIDS OVER CLICKS

Big Tech designs addiction for their profit, and our kids pay the price.

As we rapidly accelerate into the age of digital and Artificial Intelligence (AI) advancements, it's clear that our youth are bearing the brunt of the [harmful consequences](#). Big Tech profits from exploitative, dangerous algorithms that prey on vulnerable users, particularly our kids — and it's well past time to hold them accountable.



Zero Knowledge Proofs Alone Are Not a Digital ID Solution to Protecting User Privacy

BY ALEXIS HANCOCK AND PAIGE COLLINGS | JULY 25, 2025

Will anonymous credentials make it so you need an ID to use the internet? To use a computer? Is this a goal worth pursuing?